

MLPC 2026 Task 4: Data Classification

Monday, April 20, 2026

Context

Kepler Intelligent Audio Labs (KIAL), a soon-to-be-founded innovative AI startup, aims to collect an audio recording dataset from domestic environments with strong temporal annotations for a set of pre-defined target sound event classes. The goal is to capture a diverse range of sound events across varied domestic settings in order to reflect real-world acoustic conditions and enable robust sound event detection systems for different customer use cases (see Figure 1 for a schematic illustration of sound event detection).

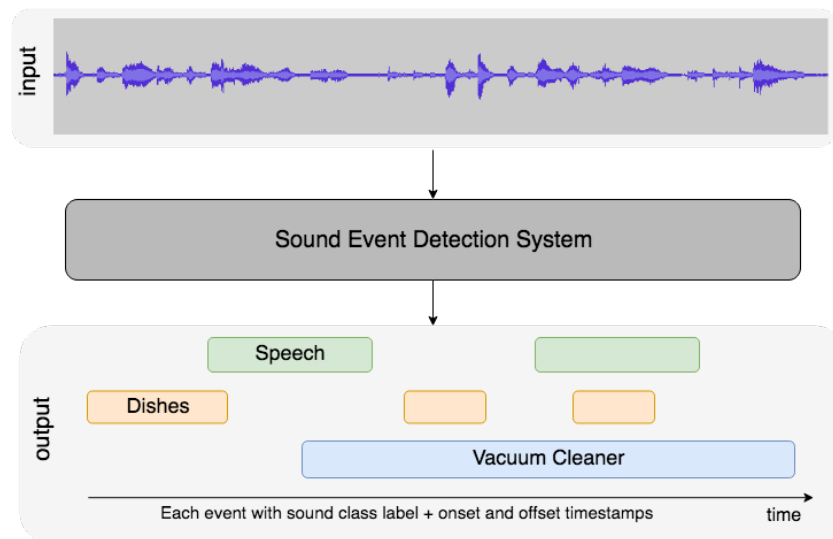


Figure 1: **Sound Event Detection (SED)** systems take an audio recording as input and predict acoustic events of interest (here: Speech, Dishes, and Vacuum Cleaner), *including* their temporal onsets and offsets. Figure taken from the DCASE website.

To realize their vision, KIAL has identified the following milestones:

1. **Data Collection:** Define the target sound event classes and collect audio recordings containing these events in representative domestic environments. (✓)
2. **Data Annotation:** Human annotators label the recordings by identifying sound events and marking their temporal onsets and offsets. (✓)
3. **Data Analysis:** Validate and explore the dataset to understand its structure, variability, and potential challenges. (✓)
4. **Classifier Training & Selection:** Train and optimize classifiers to map audio features to sound event labels and select the best-performing approach. (← we are here)
5. **Customer Validation:** Evaluate the system on a secret test set to obtain an unbiased estimate of detection performance.

Task Outline: Data Classification (40 points)

Deadline: Thursday, May 28th, 23:59

This task is mandatory. Content and formal requirements are outlined below.

Submission: Submit your team's report and slides via Moodle.

Group Work: Both team members are required to contribute to every phase of the project. Talk to your teammate early and decide on a distribution of workload, a schedule, and deadlines. *Check in often.* Plan enough time to help your colleague if they struggle to complete a task. Let us know early if your teammate is unresponsive.

Team Regrouping Policy: To allow teams to adjust collaboration if necessary, a one-time regrouping window is offered.

- **Deadline:** Friday, April 24th, 23:59
- Requests must be submitted via email to `mlpc-cp@jku.at` in the following format:

Current Team Name:

Current Team Mate:

Reason for Regrouping Request:

New Desired Team Mate (can be left empty. If not, make sure you put the other student in CC):

- Requests that are incomplete or submitted after the deadline will not be considered.
- No further team changes will be allowed after this deadline.
- One-person teams will be merged into new teams after the regrouping deadline.

1 Overview

In this task, you will use your insights from the data exploration phase to train machine learning models that predict the presence or absence of sound events for short time segments of audio recordings.

To achieve this, you will need to:

- Convert annotations into suitable training labels by aggregating them across different annotators
- Create data splits to reliably estimate your classifier's performance
- Choose a suitable metric to evaluate your classifiers
- Train classifiers from different model classes and tune their hyperparameters
- Analyze and interpret your classifiers' predictions

Hint: If your experiments become too time-consuming in terms of runtime, you may reduce their scope by selecting a smaller subset of features or subsampling the training data. Make sure to clearly state any such choices in your report.

2 Report (max. 37 points)

For the first part of this assignment, you are required to write a report based on the template provided on Moodle. Your report must address *all of the following points*:

1. **Dataset Preparation** (10 points): Prepare the dataset for classification experiments.

- (a) **Label Aggregation:** Convert the annotations from multiple annotators into training labels for each time segment. You may use your insights from the exploration phase.
 - i. Which aggregation strategy did you use?
Expected Answer: A clear description of how annotations of shape [T, C, A] are processed into labels of shape [T, C].
 - ii. Why is your aggregation strategy suitable for this classification task? What are its advantages and potential limitations?
Expected Answer: A justification of the chosen strategy, including its advantages and potential limitations compared to alternative approaches.
- (b) **Data Split:** Describe how you split the data for model selection and performance evaluation. Remember: you will need to train your classifier, tune hyperparameters, and estimate final performance.
 - i. What are the potential causes of **information leakage** when performing the split and how can you prevent it? Should audio segments from the same recording or recordings from the same collector appear in multiple splits?
Expected Answer: A discussion of potential sources of information leakage (e.g., segments from the same recording appearing in different splits) and a strategy to prevent it.
 - ii. How do you ensure that class distributions are consistent across the splits? Should the class distribution be preserved across training, validation, and test sets?
Expected Answer: A justification of whether and why class distributions should be kept similar across splits. A description of how this could be achieved. A table or visualization showing the class distributions across the splits.
- (c) **Preprocessing:** Did you apply any preprocessing techniques (e.g. data normalization, feature selection) to the audio features? If so, explain which techniques you used and why they were necessary. You may revisit and refine your preprocessing choices after selecting your model.
Expected Answer: A justified preprocessing pipeline, supported by experiments or reasoning about the chosen classifier and feature properties.

2. Evaluation (5 points):

- (a) Which evaluation criterion did you choose to compare hyperparameter settings and algorithms, and why?
Expected Answer: Choose a metric that takes into account the dataset characteristics. You may use your analysis from Task 3 to justify your choice.
- (b) What is the baseline performance? What could be the best possible performance?
Hint: A simple baseline could, for example, generate predictions based on the empirical class frequencies (e.g., by sampling according to how often each class occurs in the data), without training a classifier.
Expected Answer: Define and evaluate a simple baseline (i.e., without training a classifier). Report its performance and discuss whether perfect performance is achievable given the dataset characteristics.

3. Experiments (12 points):

- (a) Choose at least two classifiers from different model classes (Support Vector Machines, Nearest Neighbor Classifiers, Decision Trees, Neural Networks, etc). Systematically vary the most important **hyperparameters** and visualize the change in performance.
Expected Answer: For each classifier, analyze the most relevant hyperparameters and describe their role in the classifier. Systematically vary these hyperparameters, visualize how they affect performance, and clearly explain how you selected the final values.
- (b) After selecting appropriate hyperparameters, compare the final performance estimate of the two chosen classifiers to the baseline performance established in 2.b.
Expected Answer: Both classifiers should outperform the baseline. Compare the classifiers with respect to their performance and discuss their strengths and weaknesses. Explain why some classifiers may be more suitable for the given dataset.

4. Case Study and Reflection (10 points):

- (a) Find two interesting audio files that have not been used for training and qualitatively evaluate your classifier's predictions. Use the spectrogram and the sequence of predictions to visualize the classifier output. Listen to the audios and inspect the corresponding predictions of the classifier. How well does the classifier recognize the classes?
Expected Answer: A clear visualization of the predicted labels vs. the expected labels. Accompany the visualization with text explaining both correct predictions and errors made by the classifier.

- (b) Reflect on your overall classification results. What are typical failure cases? Which classes does the classifier detect reliably, and which ones are difficult to detect?

Expected Answer: A discussion of both strengths and weaknesses of the classifier. Investigate class ambiguity (e.g., using per-class confusion matrices) and use these to identify classes that are well recognized and those that are difficult to detect.

The report **must not exceed a page limit of 6 pages with at most 2000 words** (Overleaf: File → Word count). You may add figures into an appendix, however, there's no guarantee that we'll take them into account for grading.

The use of large language models (LLMs) for brainstorming, coding (e.g., implementation of down-projection or creating fancy matplotlib visualizations), or improving the clarity of your writing is explicitly allowed. Critically evaluate all outputs and include only content in your report that you understand and have verified.

All teams are required to include a “Disclosure of LLM and AI Tool Use”-section at the end of their report. This section must briefly *clarify the following points*:

- Whether or not you used any Large Language Models (LLMs) or other AI tools (image generators).
- Which models did you use, and for which tasks (e.g., brainstorming solution methods, improving writing, writing code).
- How did you verify the correctness of the LLM-generated content.

This “Disclosure of LLM and AI Tool Use”-section is mandatory for all teams (even if no tools were used) and does not count toward the page limit. Responsible use of LLMs will not affect your grade negatively. However, points will be deducted for severe cases of “AI slop,” or failing to include a “Disclosure of LLM and AI Tool Use”-section.

3 Slide Set (3 points)

The complementary slide set should present the results from your written report in a concise manner. More precisely, you will have to answer the questions corresponding to one of the sub-topics outlined in the previous section. The specific topic is determined based on the first letter of your group name, i.e. A for Team Autonomous Architectures, or V for Visionary Agents. To find your topic, determine the according letter, and find your topic in the following list:

First letter of group name	Topic
A-C	Case Study and Reflection
D-G	Experiments
H-Q	Dataset Preparation
R-Z	Evaluation

Table 1: Assignment of topics for the slides based on the first letter of your group name.

The **upper limit for the number of slides you should prepare is 4** (excluding an additional title slide that should contain your group name and the member names).

4 Development Dataset

The dataset for this task is the same data on our Moodle page used in the previous task, data exploration. The instructions on how to load the data and work with it are provided for the previous tasks in our first tutorial, and the second tutorial session on the 27th of April will provide you with more information on how to use the data for basic classification tasks.

5 Grading

Each of the four topics given in the task outline above will be evaluated according to the following criteria:

- **Thoroughness & Completeness:** Have you thought about the problem, and answered every question?
- **Clarity:** Are the ideas, features, algorithms, and results described clearly? Based on your descriptions, could the reader reconstruct your experiments?
- **Presentation:** Did you select an appropriate way of communicating your results, e.g., did you use meaningful plots where helpful?
- **Correctness:** Is the proposed procedure/experiment sound, correct?
- **Punctuality:** The reports must be submitted in time. No late submissions will be accepted.

You will be awarded all points for the slide set if it addresses the assigned topic, is within the slide limit, and is submitted before the deadline.

6 Summary

- Completing the data classification task (Task 4) is a requirement to pass this course.
- Look at the given questions and answer all of them appropriately in a written report. Make sure to use the report template provided to you via Moodle. Adhere to the given page limit (max. 6 pages with at most 2000 words).
- Add a “Disclosure of LLM and AI Tool Use”-section to your report. This section does not count toward the page limit.
- Create a set of slides tackling the questions of the topic assigned to your group. The topic is determined by the first letter of your group name. Make sure to adhere to the slide limit for this step as well (max. 4 + 1 title slide).
- Upload the written report as well as your slides to Moodle before the deadline.
- You will get a maximum number of 37 points for your written report, and 3 points for the slide set.

A The Development Dataset Descriptions

The dataset package has the following structure:

```
MLPC2026_dataset_development.zip
|-- metadata.csv
|-- annotations.csv
\-- audio_features/
    |-- 000001.npz
    |-- 000002.npz
    \-- ...
```

Raw waveforms are not required for this phase of the project but separately available in case you'd like to listen to the recordings.

```
MLPC2026_dataset_development_raw_audio.zip
|-- 000001.wav
|-- 000002.wav
\-- ...
```

A.1 metadata.csv

Contains metadata for each audio file. Columns:

Column	Description
filename	file identifier
collector_id	anonymized identifier for the student who collected the recording
target_classes	semicolon separated list of target classes present in the recording (user-provided)
non_target_classes	semicolon separated list of target classes present in the recording or None
recording_device	recording device (user-provided)
device_placement	mobile or static
recording_environment	semicolon separated list of recording locations
scene_description	textual description of the recording scene
license	license of the recording

A.2 annotations.csv

Contains annotation events in tabular form. Each row corresponds to one annotated event. Files might contain multiple annotation events. Columns:

Column	Description
filename	file identifier
annotator_id	anonymized identifier; not connectable with collector_id
annotation	selected class

<code>onset</code>	onset time in seconds
<code>offset</code>	offset time in seconds
<code>is_own_recording</code>	whether annotator equals collector

A.3 *.npz Files

Features and annotations are provided as aggregations over one second windows with a hop size of 0.5 seconds. Filenames are consistent with the filenames specified in `metadata.csv` and `annotations.csv` (e.g., `file1.wav` corresponds to `file1.npz`). Load the feature-annotation files with:

```
import numpy as np

features_and_annotations = dict(
    np.load('path/file_name.npz', allow_pickle=True)
)
```

The result is a dictionary holding the features and labels as numpy arrays:

```
{
    # features
    'mfcc_mean': ...,          # [num_segments, feature_dimensionality]
    'mfcc_std': ...,          # [num_segments, feature_dimensionality]
    ...
    # metadata
    'annotations': ...,       # [num_segments, num_classes, num_annotators]
    'start_time': ...,        # [num_segments,]
    'end_time': ...,          # [num_segments,]
    ...
}
```

Below is a detailed description of the feature files' content.

A.3.1 Acoustic Features

Base Key Name	Description	Shape
<code>zcr</code>	Zero Crossing Rate: waveform sign changes	[T,1]
<code>melspect</code>	Mel spectrogram with 128 mel frequency bands	[T,128]
<code>mfcc</code>	Mel-frequency cepstral coefficients (32 coefficients)	[T,32]
<code>mfcc_d</code>	First temporal derivative of MFCC (32 coefficients)	[T,32]
<code>mfcc_d2</code>	Second temporal derivative of MFCC (32 coeff.)	[T,32]
<code>flux</code>	Spectral flux: frame-to-frame spectral change	[T,1]
<code>flatness</code>	Spectral flatness: tonality vs. noisiness	[T,1]
<code>centroid</code>	Spectral centroid: spectral brightness	[T,1]
<code>bandwidth</code>	Spectral bandwidth: spectral spread	[T,1]
<code>contrast</code>	Spectral contrast: peak-valley contrast across bands	[T,7]
<code>rolloff_low</code>	Low spectral rolloff: low energy cutoff frequency	[T,1]
<code>rolloff_high</code>	High spectral rolloff: high energy cutoff frequency	[T,1]

energy	Signal energy	[T,1]
power	Signal power	[T,1]

Table 4: T corresponds to the length of the segment.

All features are computed for 32ms snippets and then aggregated over one second windows to reduce the temporal resolution. We used the following functions for aggregation; the suffix in the keys' names indicates the aggregation method.

- mean (***_mean**)
- standard deviation (***_std**)
- minimum (***_min**)
- maximum (***_max**)

To get a feature array use the feature's base name and the suffix, e.g.,

```
features_and_annotations["mfcc_mean"]
```

A.3.2 Temporal Alignment Arrays

start_time and **end_time** (in seconds) entries define the segment boundaries for each aggregated feature frame.

Key	Description	Shape
start_time	segment start boundaries	[T]
end_time	segment end boundaries	[T]

Table 5: T corresponds to the length of the segment.

A.3.3 Annotations

For your convenience, we've converted the annotation in **annotations.csv** into an annotation array that is aligned with the features. Each value represents proportional overlap between an annotation event and a time segment.

Key	Description	Shape
annotations	time-aligned annotation indicators for segments and annotators	[T, C, A]

Table 6: T, C, and A correspond to the length of the segment, the number of classes, and the number of annotators.

The values in **annotation** are interpreted as follows:

- 0.0: no overlap of the segment with an annotation
- 1.0: full overlap of the segment with an annotation
- values in (0,1): degree of overlap of the segment with an annotation

A.3.4 Additional Label Metadata

Additional metadata which might be useful:

Key	Description
<code>class_names</code>	list of classes; corresponds to the class indices in the annotation array; alphabetically sorted, aligned with the second dimension of the annotation array and consistent across all feature files
<code>annotator_ids</code>	list of annotator identifiers; aligned with last dimension annotation array
<code>is_own_recording</code>	list of boolean flags indicating whether each annotator is the same as the collector of the recording; aligned with last dimension of annotation array
<code>target_classes</code>	list of target class labels specified by the uploading user
<code>non_target_classes</code>	list of non-target class labels specified by the uploading user
<code>recording_device</code>	recording device; specified by the uploading user
<code>recording_environments</code>	list of rooms the recording was captured in; specified by the uploading user
<code>scene_description</code>	textual description of the audio's content; specified by the uploading user
<code>device_placement</code>	mobile or static; specified by the uploading user

A.4 License

This preliminary version of the dataset is restricted for use only within the scope of this course. The features, the metadata and the annotations may not be redistributed, shared, or published outside of this course without explicit permission from the course instructors. Audio recordings are associated with individual licenses which can be found in the metadata.csv file.